

SUBSTR: specify the positions

Make a hidden translation choice visible

01 / THE SAS EXAMPLE

```
data test;
  length text $20 first_three $3
         middle $4 from_seventh $14;
  input text :$20.;
  first_three = substr(text, 1, 3);
  middle = substr(text, 5, 4);
  from_seventh = substr(text, 7);
  datalines;
ABCDEFGHIJ
HelloWorld
SAS-to-Python
;
run;

proc print data=test noobs;
run;
```

REPORTED TEST SETUP / 21 SEP 2026
Claude Sonnet 4.5 / AWS Bedrock GovCloud
Claude Code 2.1.278 / effortLevel=high / prompt v0.1.0

WHAT YOU WILL PRACTICE

Use concrete examples to check a translation rule instead of judging code by appearance.

Extract the beginning, middle, and remainder of each string. SAS positions start at 1; Python slice positions start at 0. Check each result character by character.

EXPECTED RESULT

Row	first_three	middle	from_seventh
1	ABC	EFGH	GHIJ
2	Hel	oWor	orld
3	SAS	to-P	-Python

The table hides trailing storage blanks for readability. Stored widths are 20 for text and 3, 4, 14 for the extracted columns.

Expected values checked in Python; SAS execution pending.

REPORTED FULL MATCHES
Prompt A: 0/3 Prompt B: 3/3

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

```
Convert this SAS code to Python.  
  
Show code in this response. Do not create or run files,  
or read existing Python files.
```

REPORTED TRIAL RESULTS / 21 SEP 2026

0/3 runs matched every expected row

Run	Row 1	Row 2	Row 3	Full match
A1	Yes	No	Yes	No
A2	Yes	No	No	No
A3	Yes	No	No	No

All three chose pandas bracket slicing (.str[start:stop]).
No right-padding or assertions were reported.
A1: CSV row flags show 2/3 rows matched; the narrative summary says 1/3. This table follows the CSV.

SOURCES: results-20260921.csv; trial-log-20260921.md; analysis-20260921.md

WHY CHOICES REMAIN OPEN

- 1

The code has the rule

Start positions and lengths are in the SAS source. The model still has to translate them correctly.
- 2

The method is open

Bracket slices and .str.slice() can both be correct. A shared method does not prove shared behavior.
- 3

Appearance is a weak check

All A runs matched row 1. Checking only ABC would miss the reported failures elsewhere.

SAME INPUT, DIFFERENT REPORTED VALUES

SAS-to-Python / middle / expected: to-P

RUN A2

-to-

RUN A3

o-Py

ANTI-PATTERN: ACCEPTING A PLAUSIBLE TABLE

Execute the returned code and compare every value. Right-padding preserves widths; it does not fix a shifted start position.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS code to Python using pandas.
- Build a DataFrame with text in the supplied row order.
- Right-pad text with spaces to width 20 before slicing.
- Use .str.slice(). Subtract 1 from the SAS start position.
  Python stop = adjusted start + requested length.
- Create first_three with slice(0, 3), middle with slice(4, 8),
  and from_seventh with slice(6, None).
- Preserve result widths 3, 4, 14; trim trailing spaces only
  in the printed view, not in the stored columns.
- Include assertions for the widths and these visible triples:
  ABCDEFGHIJ: ABC, EFGH, GHIJ
  HelloWorld: Hel, oWor, orld
  SAS-to-Python: SAS, to-P, -Python
- Print all columns and briefly explain the position adjustment.

Show code in this response. Do not create or run files,
or read existing Python files.
```

SAS REFERENCE

Evidence: six fresh parallel subagents, three per prompt; raw transcripts were not supplied.
Assertions passing is reported, not independently rerun here. Python reference checked; SAS run pending.

WHY THESE DETAILS HELP

- 1 Translate the rule**
SAS start - 1 = Python start. Add the requested length to get the exclusive stop.
- 2 Separate value and display**
Preserve stored widths 20, 3, 4, 14. Trim trailing spaces only in the printed view.
- 3 Check every slice**
Assert all nine visible strings and column widths. B1-B3 report padding and passing assertions.

REPORTED RESULTS / 3 OF 3 FULL MATCHES

Run	Row 1	Row 2	Row 3	Full match
B1	Yes	Yes	Yes	Yes
B2	Yes	Yes	Yes	Yes
B3	Yes	Yes	Yes	Yes

All three used pandas .str.slice().

CHECK IT YOURSELF
Execute the code; verify values and widths.
3/3 agreement is not a guarantee of determinism.